SPARKY'S SCIENCE EXPERIMENTS

Bottle Light: Light a Dark Room Without Electricity

Physical Science (light) · Ages 7+ · 30 minutes, on a sunny day

Light up a dark room without electricity: a bottle of water in the roof spreads sunlight all around inside - a daytime light for load-shedding.

Safety first

Sparky always says: Never look straight at the sun - it can hurt your eyes!

- An adult must cut the box.

You need

- A clear 1.5 or 2-litre plastic bottle with its cap (label off)
- Clean water
- A cardboard box with a lid
- Scissors or a craft knife (adult only)
- Sticky tape

Steps

1. Fill the bottle to the top with water and close it tightly.
2. An adult cuts a round hole in the top of the box, just big enough for the body of the bottle to slide through, and a small peep-hole in one side.
3. Push the bottle into the top hole so the top part sticks out and the rest hangs inside. Tape around it so no light leaks in.
4. Put the box in bright sun and look through the peep-hole.
5. Now try an empty bottle. Which lights the box better?

If it doesn't work, try these in order

1. Not bright? The top of the bottle needs direct sunshine, not shade.
2. Box not dark inside? Tape over gaps where light leaks in.
3. Fill the bottle right to the top, with no big air bubble.

Why it works

Light bends when it goes from air into water - this is called refraction. The water and the round bottle bend and scatter the sunlight in all directions, so the whole box lights up instead of one small spot. An empty bottle lets light go mostly straight down. Bottle lights in real roofs have a little bleach added by an adult to stop green algae growing.

Did you know? Alfredo Moser, a mechanic in Brazil, invented the bottle light in 2002. In a sunny roof, one can be about as bright as a 40 to 60 watt bulb - but only in the day.

Ask Sparky: Why doesn't it work at night? Would muddy water work as well?